

HW 03 Che 311

$$2. (pvr)(r+\Delta r)\Delta\theta\Delta z|_{r+\Delta r} - (pvr)(r\Delta\theta\Delta z)|_r = \frac{\partial}{\partial r}(r pvr)(\Delta r\Delta\theta\Delta z)$$

$$(pvr)(\Delta r\Delta z)|_{\theta+\Delta\theta} - (pvr)(\Delta r\Delta z)|_{\theta} = \frac{\partial(pvr)}{\partial\theta}\Delta\theta\Delta r\Delta z$$

$$(pvr)(r\Delta\theta\Delta r)|_{z+\Delta z} - (pvr)(r\Delta\theta\Delta r)|_z = \frac{\partial(pvr)}{\partial z}r\Delta r\Delta\theta\Delta z$$

$$\frac{\partial p}{\partial t}r\Delta r\Delta\theta\Delta z + \frac{\partial}{\partial r}(r pvr)\Delta r\Delta\theta\Delta z + \frac{\partial(pvr)}{\partial\theta}\Delta\theta\Delta r\Delta z + \frac{\partial(pvr)}{\partial z}r\Delta r\Delta\theta\Delta z = 0$$

$$r\Delta r\Delta\theta\Delta z$$

$$\boxed{\frac{\partial p}{\partial t} + \frac{1}{r}\frac{\partial}{\partial r}(r pvr) + \frac{1}{r}\frac{\partial(pvr)}{\partial\theta} + \frac{\partial(pvr)}{\partial z} = 0}$$

3A.2

$$P = T_2 \Omega_i = 4\pi\mu\Omega_i^2 R^2 L \left(\frac{\lambda^2}{1-\lambda^2} \right)$$

$$L = 2.20 = \frac{40\text{ ft}}{3.281} = 12.19\text{ m}$$

$$\lambda = \frac{\frac{16}{2}}{\frac{16}{2} + 0.005} = 0.999375^2 = 0.9987511$$

$$\frac{k_2}{1-k^2} = \frac{0.9987511}{1-0.9987511} = 799.7$$

$$\mu = 5000 \text{ cP} \cdot \frac{10^{-3} \text{ kg}}{\text{m} \cdot \text{s}} = 5 \text{ kg/m} \cdot \text{s}$$

$$\Omega_i = \frac{50 \text{ rev}}{60 \text{ s}} \cdot \frac{2\pi \text{ rad}}{\text{rev}} = \frac{5\pi \text{ rad}}{3 \text{ s}}$$

$$R^2 = \left(\frac{8}{89.37} \right)^2 = 0.0412904 \text{ m}^2$$

$$P = 4\pi 5 \left(\frac{5}{3} \right)^2 (0.0412904)(12.2)(799.7) = \frac{693919.23 \text{ W}}{745.7}$$

$$\frac{930.56 \text{ HP}}{4000.2} = \boxed{0.116}$$

- 3B.1

a) $r_i = KR \quad r_o = R \quad v_\theta = v_\theta(r)$
 $\Omega_i \quad \Omega_o$

$$\mu \left(\frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r} (r v_\theta) \right) \right) = 0$$

$$\frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r} (r v_\theta) \right) = 0$$

$$\frac{1}{r} (v_\theta + r \frac{dv_\theta}{dr}) = \frac{v_\theta}{r} + \frac{dv_\theta}{dr}$$

$$\frac{d}{dr} \left(\frac{v_\theta}{r} + \frac{dv_\theta}{dr} \right) = \frac{1}{r} \frac{dv_\theta}{dr} - \frac{v_\theta}{r^2} + \frac{d^2 v_\theta}{dr^2}$$

$$\frac{d^2 v_\theta}{dr^2} + \frac{1}{r} \frac{dv_\theta}{dr} - \frac{v_\theta}{r^2} = 0$$

$$v_\theta = r^a$$

$$v'_\theta = a r^{a-1}$$

$$a(a-1)r^{a-2} + \frac{1}{r}(a r^{a-1}) - \frac{r^{a-2}}{r^2} = 0 \quad v''_\theta = a(a-1)r^{a-2}$$

$$r^{a-2}(a(a-1) + a - 1) = 0$$

$$a^2 - a + a - 1 = 0$$

$$a^2 - 1 = 0 \quad a = 1, -1 \rightarrow r^1, r^{-1}$$

$$v_\theta = Ar + \frac{B}{r}$$

B.C

$$-v_\theta(KR) = \Omega_i(KR) \rightarrow \Omega_i(KR) = A(KR) + \frac{B}{KR} \quad 1/R$$

$$-v_\theta(R) = \Omega_o R \rightarrow \Omega_o R = AR + \frac{B}{R} \quad 1/R$$

$$\Omega_i k = Ak + \frac{B}{kR^2}$$

$$\Omega_0 = A + \frac{B}{R^2} \quad \frac{B}{R^2} = C$$

$$\Omega_i k = Ak + \frac{C}{k}$$

$$\Omega_0 = A + C \rightarrow C = \Omega_0 - A$$

$$\left(Ak + \frac{\Omega_0 - A}{k} = \Omega_i k \right) k$$

$$Ak^2 + \Omega_0 - A = \Omega_i k^2$$

$$A(k^2 - 1) = \Omega_i k^2 - \Omega_0$$

$$A = \frac{\Omega_0 - \Omega_i k^2}{1 - k^2}$$

$$C = \Omega_0 - A = \Omega_0 - \frac{\Omega_0 - \Omega_i k^2}{1 - k^2} = \frac{k^2(\Omega_i - \Omega_0)}{1 - k^2}$$

$$B = CR^2 = \frac{k^2 R^2 (\Omega_i - \Omega_0)}{1 - k^2}$$

$$V_0(r) = \frac{\Omega_0 - \Omega_i k}{1 - k^2} r + \frac{k^2 R^2 (\Omega_i - \Omega_0)}{1 - k^2 \cdot r}$$

$$V_0(r) = \frac{kR}{1 - k^2} \left[(-\Omega_0 - \Omega_i k^2) \left(\frac{r}{kR} \right) + (-\Omega_i - \Omega_0) \left(\frac{kR}{r} \right) \right]$$

$$\tau_{r\theta} = \mu \left(\frac{dv_\theta}{dr} - \frac{v_\theta}{r} \right)$$

$$v_\theta = Ar + \frac{B}{r}$$

$$\frac{dv_\theta}{dr} = A - \frac{B}{r^2}$$

$$\frac{dv_\theta}{dr} - \frac{v_\theta}{r} = \left(A - \frac{B}{r^2} \right) - \left(A + \frac{B}{r^2} \right) = -\frac{2B}{r^2}$$

$$\tau_{r\theta} = \frac{-2\mu B}{r^2} \quad A = 2\pi r L \quad F = \tau_{r\theta} (2\pi r L)$$

$$T = rF = r \tau_{r\theta} (2\pi r L) = -2\pi L r^2 \frac{2\mu B}{r^2} = -4\pi \mu L B$$

$$B = \frac{k^2 R^2 (\Omega_i - \Omega_o)}{1 - k^2}$$

$$T_o = -4\pi \mu L \frac{k^2 R^2 (\Omega_i - \Omega_o)}{1 - k^2}$$

$$T_i = 4\pi \mu L \frac{k^2 R^2 (\Omega_i - \Omega_o)}{1 - k^2}$$

B. $v_\theta = v_\theta(r, \theta)$

$$v_\theta = \left(Ar + \frac{B}{r} \right) \sin \theta$$

$$\text{At } r = KR: \quad A(KR) + \frac{B}{(KR)^2} = \Omega_i (KR)$$

$$\text{At } r = R: \quad AR + \frac{B}{R^2} = \Omega_o R$$

$$AK + \frac{B}{KR^3} = \Omega_i k \quad C = \frac{C}{R^3}$$

$$A + \frac{B}{R^3} = \Omega_o$$

$$AK + \frac{C}{R^3} = \Omega_i k$$

$$A + C = \Omega_o \rightarrow C = \Omega_o - A$$

$$(KA + \frac{\Omega_o - A}{k} = \Omega_i k) k^2$$

$$AK^3 + \Omega_o - A = \Omega_i k^3$$

$$A(k^3 - 1) = \Omega_i k^3 - \Omega_o$$

$$A = \frac{\Omega_o - \Omega_i k^3}{1 - k^3}$$

$$C = \Omega_o - A = \frac{k^3(\Omega_i - \Omega_o)}{1 - k^3}$$

$$B = CR^3 = \frac{k^3 R^3 (\Omega_i - \Omega_o)}{1 - k^3}$$

$$V_\theta = \left(\frac{\Omega_0 - \Omega_i k^3}{1 - k^3} r + \frac{k^3 R^3 (\Omega_i - \Omega_0)}{(1 - k^3) r^2} \right) \sin \theta$$

$$V_\theta = \frac{kR}{1 - k^3} \left[(\Omega_0 - \Omega_i k^3) \left(\frac{r}{kR} \right) + (\Omega_i - \Omega_0) \left(\frac{kR}{r} \right)^2 \right] \sin \theta$$

$$T_{r\theta} = \mu r \frac{\partial}{\partial r} \left(\frac{V_\theta}{r} \right)$$

$$V_\theta = \left(Ar + \frac{B}{r^2} \right) \sin \theta$$

$$\frac{V_\theta}{r} = \left(A + \frac{B}{r^3} \right) \sin \theta$$

$$\frac{\partial}{\partial r} \left(\frac{V_\theta}{r} \right) = -\frac{3B}{r^4} \sin \theta$$

$$T_{r\theta} = -\frac{3\mu B}{r^3} \sin \theta$$

$$dT = (r \sin \theta) T_{r\theta} (r^2 \sin \theta d\theta d\phi)$$

$$dT = T_{r\theta} r^3 \sin^2 \theta d\theta d\phi$$

$$\int dT = \int -3\mu B \sin^3 \theta d\theta d\phi$$

$$T = -3\mu B \int_0^{2\pi} d\phi \int_0^\pi \sin^3 \theta d\theta$$

$$T = -3\mu B \cdot 2\pi \cdot \frac{4}{3}$$

$$T_o = -8\pi\mu \frac{k^3 R^3 (\Omega_i - \Omega_0)}{1 - k^3}$$

$$T_i = 8\pi\mu \frac{k^3 R^3 (\Omega_i - \Omega_0)}{1 - k^3}$$

$$\int_0^\pi \sin \theta (1 - \cos^2 \theta) d\theta$$

$$u = \cos \theta \quad du = -\sin \theta d\theta$$

$$\theta = 0$$

$$u = \cos 0 = 1$$

$$\theta = \pi \quad \cos \pi = -1$$

$$\int_1^{-1} (1 - u^2) - du$$

$$\int_{-1}^1 (1 - u^2) du = u - \frac{u^3}{3}$$

$$\left(1 - \frac{1}{3} \right) - \left(-1 + \frac{1}{3} \right) = \frac{4}{3}$$

3B.4

$$a) \quad \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 v_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (v_\theta \sin \theta) + \frac{1}{r \sin \theta} \frac{\partial v_\phi}{\partial \phi} = 0$$

$$v_r = 0 \quad v_\phi = 0$$

$$\frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (v_\theta \sin \theta) = 0 \rightarrow \frac{\partial}{\partial \theta} (v_\theta \sin \theta) = 0$$

$$\boxed{v_\theta \sin \theta = u(r)}$$

$$b) \quad 0 = -\frac{1}{r} \frac{d^2 p}{d\theta^2} + \mu \left[\frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dv_\theta}{dr} \right) \right]$$

$$v_\theta = \frac{u(r)}{\sin \theta}$$

$$\boxed{0 = -\frac{1}{r} \frac{d^2 p}{d\theta^2} + \mu \left[\frac{1}{r^2 \sin \theta} \frac{d}{dr} \left(r^2 \frac{du}{dr} \right) \right]}$$

$$c) \quad 0 = -\frac{1}{r} \frac{d^2 p}{d\theta^2} + \mu \left[\frac{1}{r^2 \sin \theta} \frac{d}{dr} \left(r^2 \frac{du}{dr} \right) \right] \cdot r \sin \theta$$

$$0 = -\sin \theta \frac{d^2 p}{d\theta^2} + \frac{\mu}{r} \frac{d}{dr} \left(r^2 \frac{du}{dr} \right)$$

$$\sin \theta \frac{d^2 p}{d\theta^2} = B \quad \frac{\mu}{r} \frac{d}{dr} \left(r^2 \frac{du}{dr} \right) = B$$

$$\frac{d^2 p}{d\theta^2} = \frac{B}{\sin \theta}$$

$$\int_{p_1}^{p_2} dp = B \int_{\epsilon}^{\pi-\epsilon} \frac{d\theta}{\sin \theta} = B \ln \frac{\tan \frac{1}{2}(\pi-\epsilon)}{\tan \frac{1}{2}\epsilon} = B \ln \cot\left(\frac{\epsilon}{2}\right) - \ln \tan\left(\frac{\epsilon}{2}\right)$$

$$= \ln \cot\left(\frac{\epsilon}{2}\right) + \ln \cos\left(\frac{\epsilon}{2}\right) = 2 \ln \cot\left(\frac{\epsilon}{2}\right)$$

$$p_2 - p_1 = 2B \ln \cot\left(\frac{\epsilon}{2}\right)$$

$$\boxed{B = \frac{p_2 - p_1}{2 \ln \cot\left(\frac{\epsilon}{2}\right)}}$$

$$\frac{u}{r} \frac{d}{dr} \left(r^2 \frac{du}{dr} \right) = B \cdot \frac{r}{u}$$

$$\int \frac{d}{dr} \left(r^2 \frac{du}{dr} \right) = \int \frac{Br}{u}$$

$$r^2 \frac{du}{dr} = \frac{B}{u} \frac{r^2}{2} + C_1$$

$$\int \frac{du}{dr} = \int \frac{B}{2u} + \frac{C_1}{r^2}$$

$$u(r) = \frac{B}{2u} r - \frac{C_1}{r} + C_2$$

$$u = \frac{BR}{2u} \left(\frac{r}{R} - \frac{C_1 R}{r} + C_2 \right) \quad \begin{matrix} u(R) = 0 \\ u(kR) = 0 \end{matrix}$$

$$0 = \frac{BR}{2u} (1 - C_1 + C_2)$$

$$1 - C_1 + C_2 = 0 \quad C_2 = C_1 - 1$$

$$r = kR$$

$$0 = \frac{BR}{2u} \left(k - \frac{C_1}{k} + C_2 \right)$$

$$k - \frac{C_1}{k} + C_2 = 0$$

$$k - \frac{C_1}{k} + C_1 - 1 = 0 \cdot k$$

$$k^2 - C_1 + kC_1 - k = 0$$

$$C_1(k-1) + k(k-1) = 0$$

$$(C_1 + k)(k-1) = 0$$

$$C_1 = -k$$

$$C_2 = -k + 1$$

$$u = \frac{p_1 - p_2}{4u \ln \cot(\epsilon/2)} R \left[\left(1 - \frac{r}{R} \right) + k \left(1 - \frac{R}{r} \right) \right]$$

$$d) \quad W = \int_0^{2\pi} \int_{KR}^R \rho u(r) r dr d\phi$$

$$W = 2\pi \rho \int_{KR}^R u(r) r dr$$

$$W = 2\pi \rho \int_{KR}^R \frac{(\rho_1 - \rho_2) R}{4\mu \ln \cot(\epsilon/2)} \left[\left(1 - \frac{r}{R}\right) + \ln\left(1 - \frac{R}{r}\right) \right] r dr$$

$$W = \frac{\pi \rho (\rho_1 - \rho_2) R}{2\mu \ln \cot(\epsilon/2)} \int_{KR}^R \left[r - \frac{r^2}{R} + Kr - KR \right] dr$$

$$= \int_{KR}^R \left((1-K)r - \frac{r^2}{R} - KR \right) dr = \left. \frac{1+K}{2} r^2 + \frac{r^3}{3R} - KRr \right|_{KR}^R$$

$$\frac{1+K}{2} R^2 - \frac{1}{3} R^2 - KR^2 = R^2 \left(\frac{1+K}{2} - \frac{1}{3} - K \right)$$

$$\frac{1+K}{2} (KR)^2 - \frac{(KR)^3}{3R} - KR(KR)$$

$$= \frac{1+K}{2} K^2 R^2 - \frac{K^3 R^2}{3} - K^2 R^2 = R^2 \left(\frac{1+K}{2} K^2 - \frac{K^3}{3} - K^2 \right)$$

$$= R^2 \left(\frac{1+K}{2} - \frac{1}{3} - K \right) - R^2 \left(\frac{1+K}{2} K^2 - \frac{K^3}{3} - K^2 \right) = \frac{R^2}{6} (1-K)^3$$

$$\boxed{W = \frac{\pi (\rho_1 - \rho_2) R^3 (1-K)^3 \rho}{12\mu \ln \cot(\epsilon/2)}}$$

Proofs

1. $\frac{\partial \rho}{\partial t} = -\operatorname{div}(\rho \mathbf{v})$

$$\frac{\partial}{\partial t}(\rho \phi) = \phi \frac{\partial \rho}{\partial t} + \rho \frac{\partial \phi}{\partial t}$$

$$\operatorname{div}(\rho \phi \mathbf{v}) = \phi \operatorname{div}(\rho \mathbf{v}) + \rho \mathbf{v} \cdot \nabla \phi$$

$$\frac{\partial}{\partial t}(\rho \phi) + \operatorname{div}(\rho \phi \mathbf{v}) = \phi \left(\frac{\partial \rho}{\partial t} + \operatorname{div}(\rho \mathbf{v}) \right) + \rho \left(\frac{\partial \phi}{\partial t} + \mathbf{v} \cdot \nabla \phi \right)$$

$$= \rho \left(\frac{\partial \phi}{\partial t} + \mathbf{v} \cdot \nabla \phi \right)$$

$$\frac{\partial \phi}{\partial t} = \frac{\partial \phi}{\partial t} + \mathbf{v} \cdot \nabla \phi$$

$$\frac{\partial}{\partial t}(\rho \phi) + \operatorname{div}(\rho \phi \mathbf{v}) = \rho \frac{\partial \phi}{\partial t}$$

If $\phi = \frac{v^2}{2}$

$$\frac{\partial}{\partial t} \left(\rho \frac{v^2}{2} \right) + \operatorname{div} \rho \frac{v^2}{2} \langle \mathbf{v} | = \rho \frac{\partial}{\partial t} \frac{v^2}{2}$$

If $\phi = |\mathbf{v}|$

$$\frac{\partial}{\partial t} \rho |\mathbf{v}| + \operatorname{div} \rho |\mathbf{v}| \langle \mathbf{v} | = \rho \frac{\partial}{\partial t} |\mathbf{v}|$$

$$2. \quad \mathbf{v} = (u, v, w)$$

$$\nabla \mathbf{v} = \begin{bmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{bmatrix}$$

$$(\nabla \mathbf{v})^T = \begin{bmatrix} \frac{\partial u}{\partial x} & \frac{\partial v}{\partial x} & \frac{\partial w}{\partial x} \\ \frac{\partial u}{\partial y} & \frac{\partial v}{\partial y} & \frac{\partial w}{\partial y} \\ \frac{\partial u}{\partial z} & \frac{\partial v}{\partial z} & \frac{\partial w}{\partial z} \end{bmatrix}$$

$$\text{div}((\nabla \mathbf{v})^T) = \begin{bmatrix} \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial x} \right) + \frac{\partial}{\partial y} \left(\frac{\partial u}{\partial y} \right) + \frac{\partial}{\partial z} \left(\frac{\partial u}{\partial z} \right) \\ \frac{\partial}{\partial x} \left(\frac{\partial v}{\partial x} \right) + \frac{\partial}{\partial y} \left(\frac{\partial v}{\partial y} \right) + \frac{\partial}{\partial z} \left(\frac{\partial v}{\partial z} \right) \\ \frac{\partial}{\partial x} \left(\frac{\partial w}{\partial x} \right) + \frac{\partial}{\partial y} \left(\frac{\partial w}{\partial y} \right) + \frac{\partial}{\partial z} \left(\frac{\partial w}{\partial z} \right) \end{bmatrix}$$

$$\text{div}((\nabla \mathbf{v})^T) = \nabla(\nabla \cdot \mathbf{v})$$

$$\frac{\partial p}{\partial t} + \text{div}(p\mathbf{v}) = 0$$

$$p = \text{constant}$$

$$p \text{div} \mathbf{v} = 0$$

$$\text{div} \mathbf{v} = 0 \rightarrow \text{div}((\nabla \mathbf{v})^T) = \nabla(\text{div} \mathbf{v}) = \nabla(0) = 0$$

$$\boxed{\text{div}((\nabla \mathbf{v})^T) = 0}$$